# Do Pruning and Quantization Compose? Order and Interaction in Post-Training Compression of Language Models

[Authors and affiliations to be confirmed]

**Abstract**. Post-training pruning and quantization are routinely stacked to compress language models, under the implicit assumption that their errors are independent. We test this assumption inside one optimal-brain-surgeon framework, comparing joint, prune-then-quantize and quantize-then-prune compression at equal storage cost on three models (135M–0.5B), and we re-analyse published 7–8B results. A decomposition of the layer error predicts sub-additivity whenever the pruning mask is chosen before the quantization grid. Experiments confirm it in all 36 mask-first cases. Quantizing first is super-additive at 4 bits and gives the worst perplexity in 16 of 18 settings, up to twice the best one.

## 1 Introduction

Pruning removes weights; quantization shortens the remaining ones. Both have accurate one-shot, post-training versions for large language models (LLMs) built on the same second-order principle: SparseGPT [1] for pruning and GPTQ [2] for quantization, both descendants of Optimal Brain Surgeon (OBS) [3, 4]. Stacking them is attractive because the gains multiply [5]. In earlier work [6] we showed that, at an equal theoretical compression rate (TCr), 25% pruning with 4-bit quantization retains about 20% more performance than pure 3-bit quantization on 7–8B models. That study, like most practice, applied the two methods one after the other and assumed that their errors do not interact.

This assumption is not innocuous. Harma et al. [7] proved that sparsity and max-scaled quantization are non-orthogonal and that sparsity should come first. Joint methods [8, 9] report that naive stacking leaves accuracy on the table. What is missing is a direct measurement, inside the Hessian-based post-training pipeline used in practice, of *how* the two errors combine and of whether the answer depends on model size.

We answer three questions. **Q1:** is the combined layer error the sum of the individual errors? **Q2:** at fixed TCr, does the order of the operations matter? **Q3:** does the interaction change with scale? Our contributions are: (i) a single OBS sweep that implements pruning, quantization and their three combinations with identical machinery, so that only the order differs; (ii) a decomposition of the combined layer error that explains when it is sub-additive; (iii) a controlled study of 93 runs on three models (135M–0.5B) on a free GPU, complemented by a re-analysis of the published 7–8B results of [6].

## 2 One framework for pruning, quantization and their combinations

For a linear layer with weights $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and calibration inputs $X$, post-training compression solves $\min_{\widehat{W}} \|WX - \widehat{W}X\|_F^2$ under a sparsity or precision constraint. With $H = 2XX^\top$ and $\Delta = W - \widehat{W}$, the objective is $\text{tr}(\Delta H \Delta^\top)$, and we report the normalized layer error $E(\widehat{W}) = \text{tr}(\Delta H \Delta^\top)/\text{tr}(WHW^\top)$. OBS-type methods process the input columns in order. When column $j$ is replaced by its compressed value, the error is propagated to the remaining columns through the inverse Hessian, so later columns compensate for earlier decisions. SparseGPT zeroes, per block of 128 columns, the weights with the smallest saliency $w^2/[H^{-1}]_{jj}^2$; GPTQ rounds each column to a min–max asymmetric grid fitted per row and per group of 128 columns. We implement both as options of one sweep and define three combinations:

- **Joint** (as in [10, 1]): one sweep; in each block the mask is selected first, the grid is fitted on the kept weights, then every column is pruned and quantized before compensation.
- **P→Q**: a pruning sweep returns $W_P$ and its mask $M$; a quantization sweep on $W_P$ keeps $M$ fixed, so pruned weights stay exactly zero while their error is still compensated.
- **Q→P**: a quantization sweep returns $W_Q$ and its grid $G$; a pruning sweep on $W_Q$ selects the mask and projects every compensated weight back onto $G$.

All three have the same storage cost. With sparsity $s$ and bit-width $b$ (16-bit baseline), $\text{TCr} = 1 - \frac{b}{16}(1-s)$ [6]. Joint and P→Q are *mask-first*: the mask is known when the grid is fitted. Q→P is *grid-first*. Layers are compressed sequentially, each calibrated on the outputs of the already-compressed previous layers.

## 3 Measuring interaction

**Layer level.** Let $E_P$, $E_Q$ and $E_{PQ}$ be the layer errors of pruning only, quantization only and a combination. The combination is *additive* if $E_{PQ} = E_P + E_Q$; we define the interaction $I_E = E_{PQ} - (E_P + E_Q)$.

**Decomposition.** Write $\langle A, B\rangle_H = \text{tr}(AHB^\top)$ and $\|A\|_H^2 = \langle A, A\rangle_H$. Consider a mask $M$ and a fixed grid $q(\cdot)$ without compensation. Let $P = W \odot M$ be the pruned weights and $R = W - q(W)$ the rounding residual, split as $R = R_M + R_{\bar{M}}$ on the pruned and kept entries. Then $\Delta_P = P$, $\Delta_Q = R$ and $\Delta_{PQ} = P + R_{\bar{M}}$, which gives, up to the common normalization,

$$I_E \;\propto\; \underbrace{-\|R_M\|_H^2}_{\text{absorption} \le 0} \;+\; \underbrace{2\langle P, R_{\bar{M}}\rangle_H - 2\langle R_M, R_{\bar{M}}\rangle_H}_{\text{cross terms}}.$$

The first term is never positive: rounding errors on weights that are pruned anyway disappear. Because $P$, $R_M$ and $R_{\bar{M}}$ have disjoint supports within each row, both cross terms vanish when $H$ is diagonal, i.e. when input features are uncorrelated. In that case $E_{PQ} \leq E_P + E_Q$: *pruning and quantization are sub-additive*. The absorption argument requires that the grid can represent the pruned state; a mask-first order fits the grid to the kept weights, whereas a grid-first order forces the compensated kept weights back onto a grid fitted before pruning, which adds rounding error outside this decomposition. We therefore expect mask-first orders to be sub-additive and the grid-first order to lose this guarantee. With OBS compensation the residuals no longer have these closed forms, so Section 5 tests the prediction.

**Model level.** With semantic retention $Sr$, the ratio of compressed to original performance summed over tasks [6], independent degradations multiply. We use $I_{Sr} = Sr_{PQ} - Sr_P\, Sr_Q$; a positive value means the combination does better than independence predicts.

## 4 Experimental setup

We compress SmolLM2-135M and -360M [11] and Qwen2.5-0.5B [12]: all linear layers of all decoder blocks; embeddings and output head stay in 16 bits. Calibration uses 128 random windows of 2048 tokens from C4 [13], damping $0.01\,\overline{\mathrm{diag}(H)}$, blocks and groups of 128 columns. The grid crosses $s \in \{25, 33.3, 50\}\%$ with $b \in \{4, 3\}$, each in the three orders, plus single-method baselines ($b = 8$ included): 25 models per network. We report WikiText-2 [14] perplexity (2048-token windows) and zero-shot accuracy on ARC-Easy [15], PIQA [16] and HellaSwag [17] (first 500 items each) with lm-eval [18]. On SmolLM2-135M, 18 extra runs repeat the key cells with two more calibration seeds. Everything runs on free T4 GPUs in 16-bit; six SmolLM2-360M runs that overflowed were repeated in 32-bit (three 16-bit controls repeated in 32-bit differ by at most 0.4% in perplexity). Unit checks run at every layer: exact mask ratio, pruned weights still zero after quantization, at most $2^b$ levels per group, and exact reproduction of the dense forward pass. Code and raw results: `https://github.com/antounyaacoub/esann2027-compose` (DOI 10.5281/zenodo.23396063).

## 5 Results

**Q1: mask-first orders are sub-additive.** Fig. 1 shows the relative interaction in all 54 combined models. Both mask-first orders are sub-additive in every one of the 36 cases: by 6–24% for the joint sweep and 3–18% for P→Q. The grid-first order Q→P is super-additive in all nine 4-bit cases (+1 to +7%), as the decomposition predicts once the grid no longer matches the kept weights. At 3 bits Q→P becomes sub-additive for a different reason: 3-bit rounding already sets 31–33% of the weights exactly to zero (15–17% at 4 bits), so a grid-first pruning of 25% only selects weights that are already zero. The resulting model

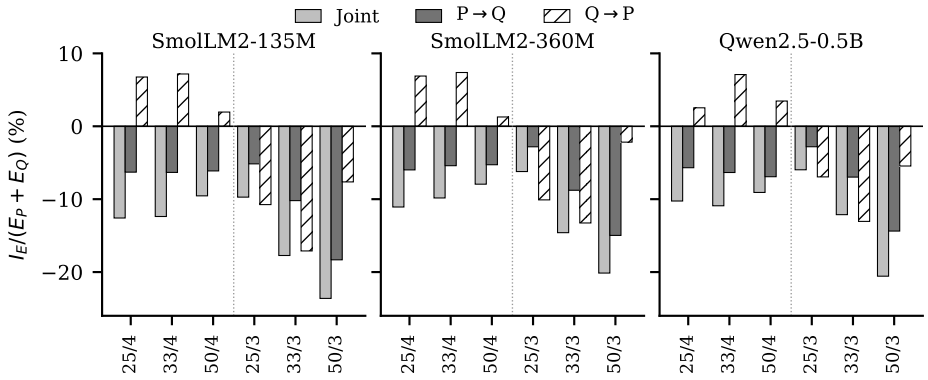



Fig. 1: Layer-level interaction $I_E$, relative to the additive prediction $E_P + E_Q$ (mean over all linear layers), for each model and each cell ($s/b$). Negative: sub-additive.

is identical to 3-bit quantization alone ($E_{PQ} = E_Q$, $I_E = -E_P$, same perplexity on all three networks): absorption in its extreme form. The TCr formula, which counts the two compressions independently, misses this free sparsity.

**Q2: the order matters at equal compression.** Table 1 compares the three orders at identical TCr. Quantizing first is the worst order in 16 of 18 settings, and its penalty grows with compression: at 50%+3-bit it reaches 1.7× (SmolLM2-135M) and 2.0× (SmolLM2-360M) the perplexity of the best order. Among mask-first orders, the joint sweep is best on SmolLM2-360M and Qwen2.5-0.5B (11 of 12 settings) and P→Q on SmolLM2-135M (5 of 6). On SmolLM2-135M at 25%+4-bit, three calibration seeds give $21.75 \pm 0.22$ for P→Q, $23.35 \pm 0.29$ for joint and $23.89 \pm 0.22$ for Q→P: the ranking is well above calibration noise. At 50%+4-bit the joint and P→Q intervals overlap ($42.6 \pm 2.6$ vs. $41.3 \pm 1.8$), while Q→P stays worse ($47.6 \pm 2.0$). The joint sweep always has the smallest layer error on 4-bit cells, yet on SmolLM2-135M it does not give the smallest perplexity: a smaller local objective does not guarantee a smaller end-to-end loss. Finally, at 3 bits a mask-first combination can beat quantization alone at higher compression: on SmolLM2-360M, 25% pruning + 3-bit gives 20.11 against 25.92 for 3-bit alone. A plausible reading is that the zeros which 3-bit rounding creates anyway are better chosen by OBS saliency than by rounding.

**Q3: interaction across scale.** The order effect shrinks with model size. At 4 bits, the spread between the best and worst order is 10–24% on SmolLM2-135M, 9–14% on SmolLM2-360M and 2–5% on Qwen2.5-0.5B; the last model also differs in architecture and training, so this trend is suggestive rather than causal. Table 2 reports the model-level interaction of the sequential pipeline. At 25%+4-bit, $I_{Sr}$ stays at or below +0.012 from 135M to 7B, inside the noise of 500-item task subsets ($\approx \pm 0.02$): at moderate compression the two methods compose multiplicatively at every scale we observe. At 33%+3-bit the small

| Model | $s + b$ | TCr | $P$ only | $Q$ only | Joint | P→Q | Q→P |
|---|---|---|---|---|---|---|---|
| SmolLM2-135M | 25% + 4 | 0.812 | 16.48 | 21.77 | 23.04 | **21.86** | 24.03 |
| (dense 15.42) | 33% + 4 | 0.833 | 18.39 | 21.77 | 26.08 | **24.11** | 27.73 |
| | 50% + 4 | 0.875 | 32.89 | 21.77 | 45.45 | **40.15** | 49.68 |
| | 25% + 3 | 0.859 | 16.48 | 58.17 | 60.87 | **56.88** | 58.17 |
| | 33% + 3 | 0.875 | 18.39 | 58.17 | 63.18 | **58.97** | 68.00 |
| | 50% + 3 | 0.906 | 32.89 | 58.17 | **88.24** | 97.97 | 150.10 |
| SmolLM2-360M | 25% + 4 | 0.812 | 12.07 | 13.29 | **13.17** | 13.32 | 14.34 |
| (dense 11.46) | 33% + 4 | 0.833 | 12.86 | 13.29 | **14.12** | 14.20* | 15.53 |
| | 50% + 4 | 0.875 | 18.64 | 13.29 | **20.82** | 21.02 | 23.82 |
| | 25% + 3 | 0.859 | 12.07 | 25.92 | **20.11*** | 20.77 | 25.92 |
| | 33% + 3 | 0.875 | 12.86 | 25.92 | **20.34*** | 20.88 | 30.50* |
| | 50% + 3 | 0.906 | 18.64 | 25.92 | **28.71*** | 30.07* | 57.80 |
| Qwen2.5-0.5B | 25% + 4 | 0.812 | 13.63 | 14.24 | **14.68** | 14.71 | 15.02 |
| (dense 13.07) | 33% + 4 | 0.833 | 14.52 | 14.24 | **15.62** | 15.69 | 15.99 |
| | 50% + 4 | 0.875 | 20.21 | 14.24 | **21.61** | 22.44 | 22.74 |
| | 25% + 3 | 0.859 | 13.63 | 21.95 | 21.96 | 22.11 | **21.95** |
| | 33% + 3 | 0.875 | 14.52 | 21.95 | **22.00** | 22.33 | 22.96 |
| | 50% + 3 | 0.906 | 20.21 | 21.95 | **28.41** | 29.69 | 35.61 |

Table 1: WikiText-2 perplexity at equal TCr (best order in bold). Single-method columns use the same $s$ or $b$ alone. *Re-run in 32-bit after a 16-bit overflow.

| Model | 25% + 4-bit | 33% + 3-bit | Tasks |
|---|---|---|---|
| SmolLM2-135M | +0.012 | +0.053 | ARC-E, PIQA, HellaSwag |
| SmolLM2-360M | +0.009 | +0.089 | ARC-E, PIQA, HellaSwag |
| Qwen2.5-0.5B | +0.009 | +0.012 | ARC-E, PIQA, HellaSwag |
| Mistral-7B-v0.3 | +0.009 | −0.005 | MMLU-Pro, BBH, MATH [6] |
| LLaMA-3.1-8B | +0.122 (+0.023) | +0.038 (+0.050) | MMLU-Pro, BBH, MATH [6] |

Table 2: Model-level interaction $I_{Sr} = Sr_{PQ} - Sr_P Sr_Q$ for prune-then-quantize. For 7–8B, recomputed from the published scores; in brackets, without MATH.

models gain from the combination ($I_{Sr}$ up to +0.09), while Mistral-7B stays at independence. LLaMA-3.1-8B shows a large positive value at 25%+4-bit, but it is driven by the anomalous collapse of 4-bit LLaMA on MATH (0.2%) [6]; without MATH it falls to +0.02.

## 6 Conclusion

Inside the second-order post-training pipeline, pruning and quantization compose better than independence predicts, provided the mask is chosen before the quantization grid. A simple decomposition explains why: rounding errors on pruned weights are absorbed, and the remaining cross terms vanish for uncorrelated inputs. All 36 mask-first combinations confirm it. Quantizing first breaks this guarantee and is the worst order in 16 of 18 settings, up to twice the best perplexity. In practice: never quantize first; prefer the joint sweep from a few hundred million parameters up. Two further observations deserve follow-up:

low-bit rounding creates a large share of free zeros that TCr ignores, and the smallest layer error does not always give the best model. Extending the analysis to compensated sweeps, semi-structured sparsity and larger models is left for future work.

## References


[1] E. Frantar and D. Alistarh. SparseGPT: Massive language models can be accurately pruned in one-shot. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 10323–10337, 2023.

[2] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations (ICLR)*, 2023.

[3] Y. LeCun, J. S. Denker, and S. A. Solla. Optimal brain damage. In *Advances in Neural Information Processing Systems 2*, pages 598–605, 1990.

[4] B. Hassibi and D. G. Stork. Second order derivatives for network pruning: Optimal Brain Surgeon. In *Advances in Neural Information Processing Systems 5*, pages 164–171, 1993.

[5] S. Han, H. Mao, and W. J. Dally. Deep compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding. In *International Conference on Learning Representations (ICLR)*, 2016.

[6] S. Laborde, M. Cousseau, A. Yaacoub, and L. Prevost. Semantic retention and extreme compression in LLMs: Can we have both? In *International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2025.

[7] S. B. Harma, A. Chakraborty, E. Kostenok, D. Mishin, D. Ha, B. Falsafi, M. Jaggi, M. Liu, Y. Oh, S. Subramanian, and A. Yazdanbakhsh. Effective interplay between sparsity and quantization: From theory to practice. In *International Conference on Learning Representations (ICLR)*, 2025.

[8] J. Guo et al. Compressing large language models by joint sparsification and quantization. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, 2024.

[9] H. Guo, Y. Li, and L. Benini. Optimal brain restoration for joint quantization and sparsification of LLMs, 2025. arXiv:2509.11177.

[10] E. Frantar, S. P. Singh, and D. Alistarh. Optimal brain compression: A framework for accurate post-training quantization and pruning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.

[11] L. Ben Allal et al. SmolLM2: When smol goes big – data-centric training of a small language model, 2025. arXiv:2502.02737.

[12] Qwen Team. Qwen2.5 technical report, 2024. arXiv:2412.15115.

[13] C. Raffel et al. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

[14] S. Merity, C. Xiong, J. Bradbury, and R. Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations (ICLR)*, 2017.

[15] P. Clark et al. Think you have solved question answering? Try ARC, the AI2 reasoning challenge, 2018. arXiv:1803.05457.

[16] Y. Bisk, R. Zellers, R. Le Bras, J. Gao, and Y. Choi. PIQA: Reasoning about physical commonsense in natural language. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 7432–7439, 2020.

[17] R. Zellers, A. Holtzman, Y. Bisk, A. Farhadi, and Y. Choi. HellaSwag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4791–4800, 2019.

[18] L. Gao et al. A framework for few-shot language model evaluation. Zenodo, version 0.4, 2024.